MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU -UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

(2024 Admission onwards)

Skill Enhancement Course- MG4SECBCH200- Biochemical Tests in Disease Diagnosis

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
(Answer all questions; each question carries 1marks)

1. Which sugar load is commonly used in a standard GCT? [K][2]
   A) 25g B) 50 g C) 75 g D) 100 g

2. Why is proper sample collection and handling critical in biochemical tests? [K][1]
   A) It reduces laboratory workload B) It ensures reliable and accurate test results
   C) It lowers the cost of testing D) It helps in faster diagnosis only

3. Elevated TSH with low T4 typically indicates: [U][3]
   A) Primary hypothyroidism B) Hyperthyroidism C) Secondary hypothyroidism D) Normal thyroid function

4. A low PCV value generally indicates [U][1]
   A) Dehydration B) Anemia C) Polycythemia D) Increased oxygen-carrying capacity

5. Which biochemical technique enables precise identification of biomolecules? [U][4]
   A) CRISPR B) Biosensors C) Mass Spectrometry D) Point of Care testing

(1x 5= 5 marks)

Part B
Fill in the Banks
(Answer all questions; each question carries 2 marks)

6. The primary action of insulin is to _____ blood glucose. [K][2]

7. _____ is the anticoagulant commonly used for ESR measurement. [U][1]

8. Serum _____ is used to assess the synthetic function of the liver [U][3]

9. Nanotechnology have ability to detect _____ concentrations of biomolecules [K][4]

10. Polycythemia is characterized by _____ hemoglobin levels. [U][1]

(2 x 5=10 marks)

Short Answer Questions
(Answer any 5 questions; each question carries 3 marks)

11. Explain why first-morning urine is preferred for certain biochemical tests. [U][1]

12. Analyze the clinical significance of RBC count in anemia. [A][1]

13. What is the Glucose Tolerance Test (GTT)? State its indication and mention the normal blood glucose values during the test. [A][2]

14. Explain the significance of HbA1c in monitoring diabetes mellitus. [U][2]

15. Name the parameters measured in thyroid function tests and mention their clinical [U[3]

significance

16. What are renal function tests? state their significance. [An][3]

17. Name any three emerging techniques and briefly mention their applications. [U][4]

(3 x 5=15 marks)

Part D
Short Essay Type Question
(Answer any four questions; each question carries 5 marks)

18. Evaluate the impact of emerging biochemical testing technologies on disease diagnosis and patient care. [E][4]

19. List the enzymes commonly measured in liver function tests and their clinical significance. [U][3]

20 Describe how you would collect and handle a blood sample for routine biochemical analysis [A][1]

21. Explain the purpose and significance of different tests used in diabetes mellitus. [U][2]

22. Analyze how renal function test results reflect kidney function in different renal disorders [An][3]

23. Give a detailed account on glucose challenge test. [U][2]

(4 x 5 = 20 marks)